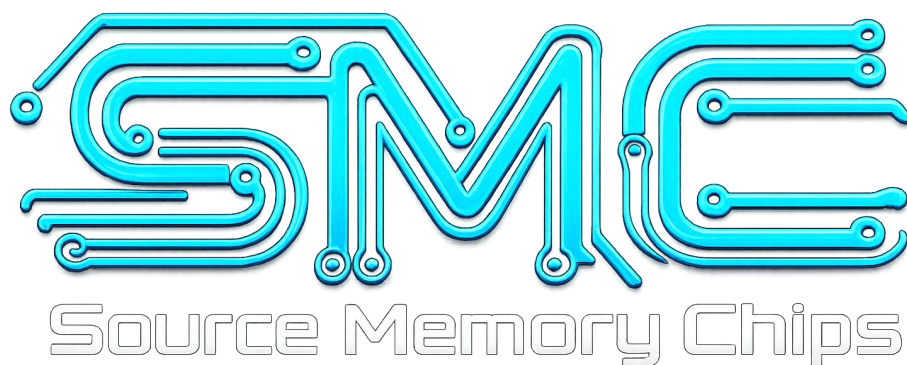


H9HP27ADAMAD

eMCP Memory Solution

Mobile / Embedded Memory Solution

Technical Datasheet

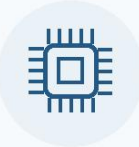


Source Memory Chips (SMC) provides technical reference documentation for memory products used in industrial, networking, embedded and long-life electronic applications.

<https://www.sourcememorychips.com>

SK hynix eMCP Memory Solution Product Specification

Product Highlights

 High Integration 32GB eMMC 5.1 + 24Gb LPDDR4X in a single package	 High Speed HS400 storage interface LPDDR4X up to 3733 Mbps
 Compact Package 254-ball FBGA 11.5 × 13.0 mm	 Embedded Ready Optimized for mobile, IoT, networking and embedded systems

1. Product Overview

The SK hynix H9HP27ADAMAD is an embedded multi-chip package (eMCP) memory solution integrating 32GB of eMMC 5.1 storage and 24Gb of LPDDR4X SDRAM in a compact 254-ball FBGA package.

The single-package architecture combines managed non-volatile storage with high-speed system memory, reducing PCB area, simplifying signal routing and supporting compact mobile and embedded platform designs.

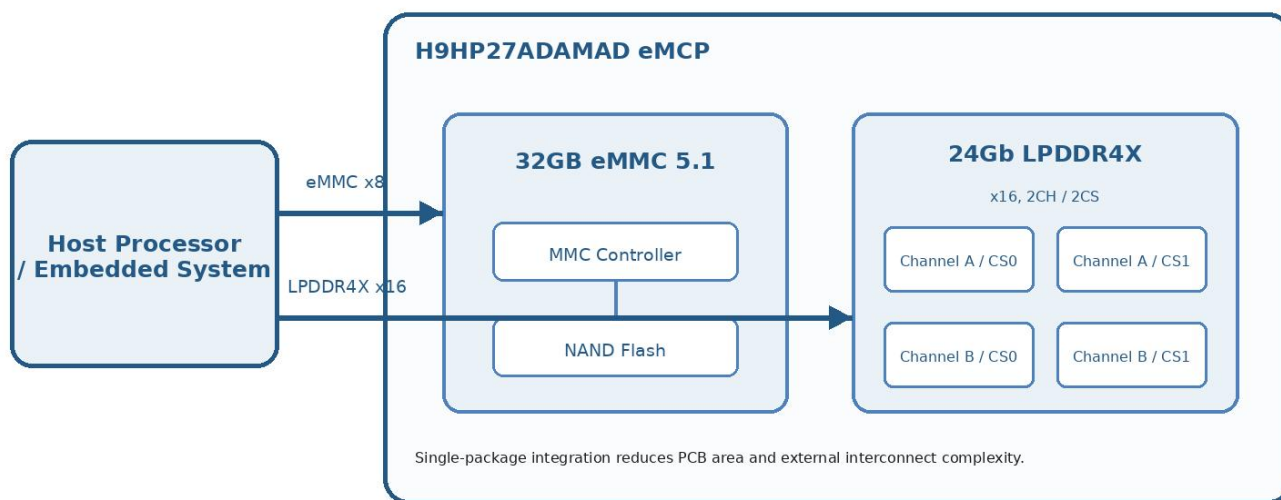
Item	Specification
Manufacturer	SK hynix
Content Model	H9HP27ADAMAD
Full Ordering Part	H9HP27ADAMADAR-KMM
Product Type	eMCP
Memory Configuration	32GB eMMC 5.1 + 24Gb LPDDR4X
Package	254-ball FBGA
Operating Temperature	-25°C to +85°C

2. Memory Architecture

H9HP27ADAMAD integrates eMMC 5.1 storage and LPDDR4X working memory within one embedded package. The host accesses the storage subsystem through the eMMC interface and the DRAM subsystem through independent LPDDR4X channels.

Memory Block	Function
32GB eMMC 5.1	Managed non-volatile storage for operating system, firmware, applications and user data.
24Gb LPDDR4X	High-speed volatile memory for application processing and multitasking.

H9HP27ADAMAD Simplified Functional Architecture



3. Product Advantages

- Compact storage and DRAM integration in one package
- Reduced PCB footprint and component count
- Simplified board routing and memory subsystem design
- HS400 eMMC and LPDDR4X high-speed interface support
- Suitable for mobile, embedded, IoT and networking platforms

4. General Specifications

Parameter	Specification
eMMC Capacity	32GB, x8 data organization
eMMC Standard	JEDEC eMMC 5.1; backward compatible with eMMC 4.5 and 5.0
eMMC Bus Width	1-bit default, 4-bit and 8-bit modes
eMMC High-Speed Modes	HS400 and HS200
LPDDR4X Capacity	24Gb, approximately 3GB
LPDDR4X Organization	x16, 2 channels / 2 chip selects
LPDDR4X Data Rate	Up to 3733 Mbps
Package	254-ball FBGA, 11.5 × 13.0 mm, 1.0 mm max
Environmental Status	Lead-free / Halogen-free / RoHS

Core Configuration

 32GB eMMC 5.1 Managed embedded storage	 24Gb LPDDR4X Approx. 3GB system memory	 3733 Mbps LPDDR4X data rate	 254-ball FBGA 11.5 × 13.0 mm package
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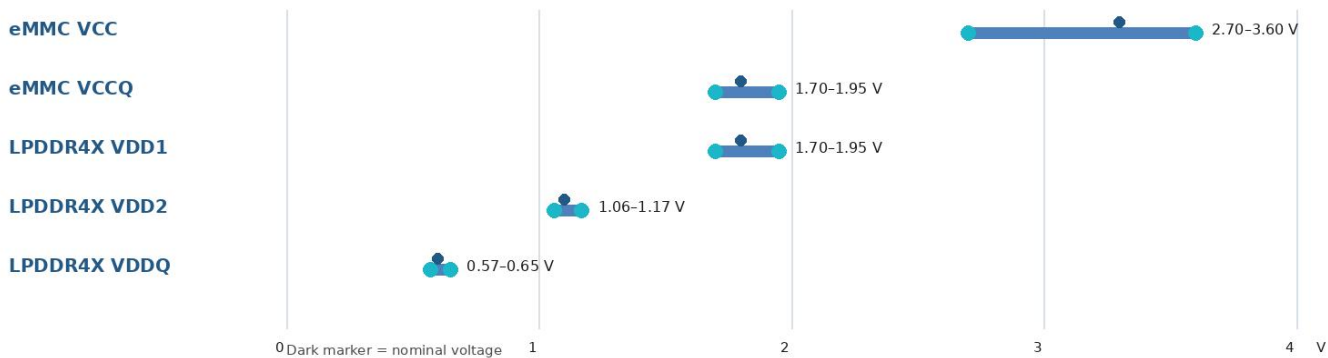
eMMC 5.1 Feature Support

Category	Supported Functions
Performance	HS400, HS200, enhanced data strobe and configurable driver strength
Data Management	Cache, discard, trim, erase and sanitize
Reliability	Reliable write, reset, health report and field firmware update
Security / Partitioning	Partitioning, RPMB, write protection and secure removal
Power Management	Sleep/awake and power-off notification

5. Electrical Characteristics & Performance

Separate power rails are used for the eMMC subsystem and LPDDR4X memory. The range chart below summarizes the operating windows; final power sequencing and decoupling should follow the official manufacturer specification.

Operating Voltage Range Overview



Performance Highlights

eMMC HS400

400 MB/s

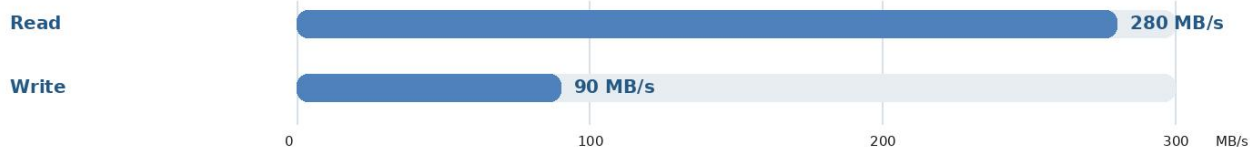
Maximum bus transfer rate

LPDDR4X

3733 Mbps

Maximum data rate

eMMC Sequential Performance — Reference Test

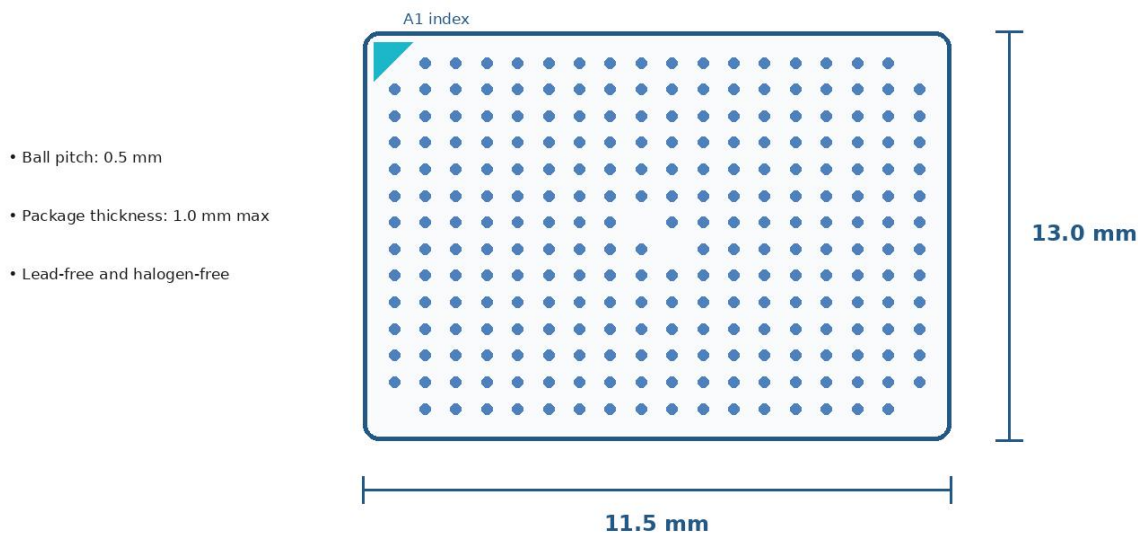


The 280 MB/s read and 90 MB/s write values are reference results under the stated vendor test conditions. Actual performance depends on host controller, firmware, workload and board design.

6. Package Information

H9HP27ADAMAD is supplied in a compact 254-ball FBGA package intended for high-density mobile and embedded layouts.

254-ball FBGA Package — Simplified Mechanical View



Mechanical Item	Specification
Package Type	254-ball FBGA
Body Size	11.5 mm × 13.0 mm
Maximum Thickness	1.0 mm
Ball Pitch	0.5 mm
Package Material	Lead-free and halogen-free

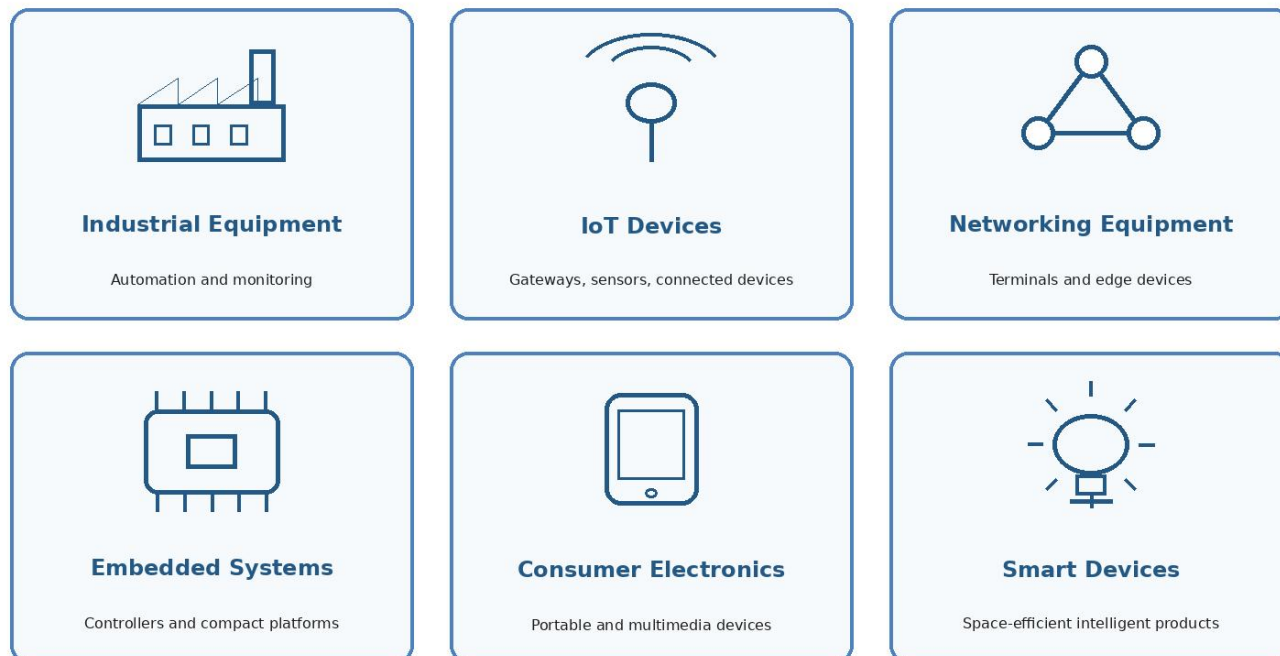
PCB and Assembly Notes

- Use the A1 index and official ball map to confirm component orientation.
- Do not derive the PCB land pattern solely from the simplified illustration.
- Confirm solder-mask definition, reflow profile and inspection criteria with the assembler.
- Use the official manufacturer ball-assignment table for all eMMC, LPDDR4X, power and ground connections.

7. Applications

The H9HP27ADAMAD eMCP solution is intended for compact systems requiring integrated storage, high-speed working memory and efficient PCB utilization.

Target Application Areas



Application Summary

Application Area	Memory Requirement	H9HP27ADAMAD Advantage
Industrial / Embedded Systems	Reliable integrated memory	Managed eMMC + LPDDR4X architecture
IoT and Edge Devices	Compact, low-power design	Space-saving eMCP package
Networking Equipment	Stable storage and fast data access	High integration capability
Smart Electronics	Optimized PCB layout	Single-package memory solution

8. Ordering Information & Disclaimer

Ordering Information

Part Number	Manufacturer	Product Type	Package
H9HP27ADAMADAR-KMM	SK hynix	32GB eMMC + 24Gb LPDDR4X eMCP	254-ball FBGA

Product Identification

Item	Description
Content Model	H9HP27ADAMAD
Product Family	Embedded Multi-Chip Package (eMCP)
Memory Configuration	32GB eMMC 5.1 + 24Gb LPDDR4X
Target Market	Mobile / Embedded Applications

Revision History

Revision	Date	Description
Rev 1.0	July 2026	Initial SMC technical-reference datasheet release

Disclaimer

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Specifications, features, ordering codes and availability may change without notice. Customers must consult the official manufacturer documentation and verify the complete device marking before final design validation, procurement or production use.

Primary technical basis: SK hynix H9HP27ADAMADAR eMCP Specification, Rev. 1.0, Mar. 2019.

Contact Information

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